

Python Companion Document

Python Installation

There are many ways to install Python. The easiest is likely to install the kernel via Python's website and use Microsoft's VSCode. Hyperlinks will direct to download pages. If you'd prefer a more integrated way, Anaconda is an all in one method which provides the kernel, an IDE, as well as a JupyterNotebook client (may be useful in future).

Package Installation

There are a few packages which will need to be installed to run my code. User will need to execute the code in the command line terminal in an IDE. A code chunk below will list an installation snippet:

```
pip install pandas
pip install numpy
pip install scipy
pip install statsmodels.api
```

- Pandas: **Pandas** is a library used for data manipulation. In the PS1 files, I mainly use it to create **DataFrame** objects to store my results in as presentation is much cleaner. Later, it will likely become more integrated into my files as complexity increases
- Numpy: **numpy** is a library used for linear algebra and numeric operations. In this problem set, you will see **np** often. This is an alias for **numpy**.
- Scipy: **scipy** is a library for optimization methods in this context. It is primarily used for the **minimize** function, similar to **fminunc** in MatLab.
- Statsmodels: **statsmodels.api** is used for regression operations. I use it in my translation of the Problem 1 code, so it is not explicitly needed to run Problem 2. However, if interested in seeing said translation, it will be needed.